5.1 Conditioning: changing the denominator

Definition and results

For $P(A) > 0$, the probability of B given A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table of equally likely counts, it is the quotient of the count of $A \cap B$ by the count of A. In general $P_A(B) \neq P_B(A)$.

Status and justification. Definition. Multiplying by $P(A)$ gives the rule $P(A \cap B) = P(A)P_A(B)$. The condition $P(A) > 0$ is essential.

Worked example. Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \dfrac{12}{40} = 0.3$, whereas $P_B(A) = \dfrac{12}{30} = 0.4$. $P(A \cap B) = 0.12$.

Watch out

"Among the A's" fixes the denominator. A conditional probability is not an intersection.

Immediate application. Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

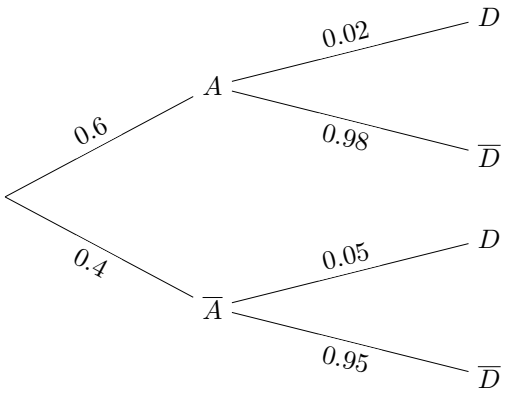
5.2 Trees, partitions and total probability

Definition and results

A partition $A_1, \ldots, A_k$ is a family of disjoint events whose union is the sample space. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. On a tree, the probabilities leaving the same node sum to 1; the probabilities along a path are multiplied; disjoint paths are combined by adding.

Status and justification. B is the disjoint union of the sets $B \cap A_i$. By additivity and by the definition of conditional probability, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(A_i)P_{A_i}(B)$.

Worked example. Two machines supply 60 % and 40 % of a production; their defect rates are 2 % and 5 %. $P(D) = 0.6 \times 0.02 + 0.4 \times 0.05 = 0.032$. Among the defective items, the first machine accounts for $\dfrac{0.012}{0.032} = \dfrac{3}{8}$.



A denotes the machine that supplies 60 percent of the production and D a defect. For example, $0.02 = P_A(D)$: the second-level probabilities are conditional.

Watch out

The simple average of the rates is correct only if the branches have equal weights.

Immediate application. $P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

5.3 Independence and disjointness

Definition and results

A and B are independent if $P(A \cap B) = P(A)P(B)$. If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. They are disjoint if $A \cap B = \varnothing$. Two disjoint events of strictly positive probability are not independent.

Status and justification. The conditional criterion follows from a division by $P(A) > 0$. For disjoint events of positive probability, the left-hand side is 0 and the right-hand product is strictly positive.

Worked example. With a fair die, let A: the result is even, B: the result is at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: independent. A and "the result is odd" are disjoint but not independent.

Watch out

Independence in a model must be justified or stated explicitly; "two different events" is not enough.

Immediate application. Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent?

5.4 Successive trials and Bernoulli trials

Definition and results

A Bernoulli trial has two outcomes: success, with probability p, and failure, with probability $1-p$. In a repetition of n independent, identical trials, with $n \leq 4$ here, the branch probabilities remain p and $1-p$. A path with k successes has probability $p^k(1-p)^{n-k}$; we count the paths that realise the event, without any general theory of the binomial distribution.

Status and justification. The probability of a path is a product, by independence. The paths corresponding to an event are disjoint: their probabilities add.

Worked example. For three trials with $p = \frac{1}{2}$, exactly two successes correspond to SSF, SFS, FSS. Each path has probability $\frac{1}{8}$, so the total is $\frac{3}{8}$. At least one success has probability $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

Watch out

When drawing without replacement from an urn, the probabilities generally change after the first draw: the independent model may be wrong.

Immediate application. Four independent trials, each with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success.

5.5 Simulating and discussing the model

Definition and results

A pseudo-random simulation produces frequencies that fluctuate around a probability of the model. For a uniform point in the square $[0;1]^2$, the proportion satisfying $y \leq x^2$ estimates the area under the parabola; the proportion satisfying $x^2 + y^2 \leq 1$, multiplied by 4, estimates π. Repetitions must use new draws.

Status and justification. Simulation principle accepted and explored experimentally. The equality of probability and area is an interpretation of the uniform draw; integration is not a prerequisite. A simulation does not prove an exact value.

Worked example. Among 10000 points, if 7840 lie in the quarter disc, the estimate of π is $\frac{4 \times 7840}{10000} = 3.136$. Two trials of the same size generally give different values.

Watch out

Fixing a seed makes a trial reproducible; it neither removes the randomness of the model nor guarantees any accuracy.

Immediate application. A simulation of 2000 points gives 675 points below $y = x^2$. What area does it estimate?